

Quantum States, Geometry, and Transport

Complete Student Lecture Handout — Lecture 1

Rigorous Mathematical Treatment with Full Derivations

Prepared by **Machiraju Karthikeya**

Quantum Computing & Photonics
PES University, Bengaluru

April 3, 2026

Learning Objectives: By the end of this lecture, you should understand:

- The complete mathematical structure of Hilbert space and why each property is physically necessary
- How inner products and the Born rule connect geometry to probability through rigorous derivation
- Why superposition, measurement, and uncertainty emerge inevitably from linear algebra alone
- The complete mathematical origin of energy quantization in confined systems with full proofs
- How transport emerges from allowed quantum modes and density of states calculations
- The tensor product structure and mathematical foundations of entanglement

Key Mathematics: Hilbert spaces, inner products, eigenvalues, operators, commutators, tensor products, spectral decomposition, functional analysis foundations

Prerequisites: Linear algebra (vector spaces, matrices, eigenvalues), complex analysis (complex numbers, complex functions)

Contents

1 Hilbert Space: The Complete Mathematical Foundation	4
1.1 Formal Definition and Axioms	4
1.2 The Induced Norm	5
1.3 The Cauchy-Schwarz Inequality	6
1.4 Why Complex Numbers Are Essential	7
1.5 Examples of Hilbert Spaces	8
1.5.1 Finite-Dimensional: Spin-1/2 System	8
1.5.2 Infinite-Dimensional: Particle in 1D Space	9
1.5.3 Three-Dimensional Configuration Space	9
2 Operators and Observables: Mathematical Formalism	10
2.1 Linear Operators	10
2.2 The Adjoint (Conjugate Transpose) Operator	10
2.3 Hermitian Operators	11
2.4 Spectral Decomposition	13
3 The Born Rule and Inner Products: From Geometry to Probability	14
3.1 The Born Rule (Fundamental Postulate)	14
3.2 Geometric Interpretation	14
3.3 Why Magnitude Squared?	15
3.4 Proof: Total Probability Equals One	16
3.5 Expectation Values and Uncertainties	17
4 Superposition and Interference: Quantum vs Classical Probability	18
4.1 The Superposition Principle	18
4.2 Probability Amplitudes vs Probabilities	19
4.3 Interference and the Double Slit Experiment	20
4.4 Relative Phase and Interference	21
5 Commutators and the Uncertainty Principle	21
5.1 Commutator Relations	21
5.2 The Canonical Commutation Relation	22
5.3 The General Uncertainty Principle	22
5.4 Position-Momentum Uncertainty	24
6 Quantum Confinement and Energy Quantization	24
6.1 The Infinite Potential Well (Particle in a Box)	24
6.2 Solving the Time-Independent Schrödinger Equation	25
6.3 Normalization and Orthonormality	26
6.4 Physical Interpretation: Quantization from Geometry	27
6.5 Two-Dimensional Well	28
7 Density of States and Transport Properties	28
7.1 Definition of Density of States	28
7.2 Calculation for One-Dimensional System	29
7.3 Calculation for Two-Dimensional System	30
7.4 Three-Dimensional System	30
7.5 Dimensional Dependence and Universality	31

8	Tensor Products and Entanglement	31
8.1	Hilbert Space for Composite Systems	31
8.2	Product States	32
8.3	Entangled States	32
8.4	Schmidt Decomposition	34
8.5	Entanglement Entropy	34
9	Summary: Mathematical Framework of Quantum Mechanics	35

1 Hilbert Space: The Complete Mathematical Foundation

1.1 Formal Definition and Axioms

Definition A complex Hilbert space $\mathcal{H}$ is a complete vector space over $\mathbb{C}$ equipped with an inner product $\langle \cdot | \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ satisfying four fundamental properties:

(1) **Sesquilinearity (linear in second argument, conjugate-linear in first):**

$$\langle \psi | \alpha \phi + \beta \chi \rangle = \alpha \langle \psi | \phi \rangle + \beta \langle \psi | \chi \rangle \quad (1)$$

$$\langle \alpha \psi + \beta \phi | \chi \rangle = \alpha^* \langle \psi | \chi \rangle + \beta^* \langle \phi | \chi \rangle \quad (2)$$

(2) **Conjugate Symmetry (Hermitian):**

$$\langle \psi | \phi \rangle = \langle \phi | \psi \rangle^* \quad \forall |\psi\rangle, |\phi\rangle \in \mathcal{H} \quad (3)$$

(3) **Positive Definiteness:**

$$\langle \psi | \psi \rangle \geq 0 \quad \text{with equality if and only if} \quad |\psi\rangle = 0 \quad (4)$$

(4) **Completeness (Cauchy Completeness):**

Every Cauchy sequence $\{|\psi_n\rangle\}$ in $\mathcal{H}$ (satisfying $\forall \epsilon > 0, \exists N : \|\psi_m - \psi_n\| < \epsilon$ for $m, n > N$) converges to an element in $\mathcal{H}$.

1.2 The Induced Norm

Definition The inner product naturally induces a norm on $\mathcal{H}$:

$$\|\psi\| := \sqrt{\langle \psi | \psi \rangle} \quad \forall |\psi\rangle \in \mathcal{H} \quad (5)$$

This norm satisfies four properties:

(1) Positive Definiteness:

$$\|\psi\| \geq 0, \quad \text{equality iff } |\psi\rangle = 0$$

(2) Homogeneity (Positive Scalability):

$$\|\alpha\psi\| = |\alpha|\|\psi\| \quad \forall \alpha \in \mathbb{C}, |\psi\rangle \in \mathcal{H}$$

(3) Triangle Inequality (Subadditivity):

$$\|\psi + \phi\| \leq \|\psi\| + \|\phi\| \quad \forall |\psi\rangle, |\phi\rangle \in \mathcal{H}$$

(4) Parallelogram Law:

$$\|\psi + \phi\|^2 + \|\psi - \phi\|^2 = 2(\|\psi\|^2 + \|\phi\|^2)$$

Derivation/Proof Proof of Parallelogram Law:

$$\|\psi + \phi\|^2 = \langle \psi + \phi | \psi + \phi \rangle = \|\psi\|^2 + \langle \psi | \phi \rangle + \langle \phi | \psi \rangle + \|\phi\|^2 \quad (6)$$

$$= \|\psi\|^2 + \langle \psi | \phi \rangle + \langle \psi | \phi \rangle^* + \|\phi\|^2 \quad (7)$$

$$= \|\psi\|^2 + 2\text{Re}(\langle \psi | \phi \rangle) + \|\phi\|^2 \quad (8)$$

Similarly:

$$\|\psi - \phi\|^2 = \|\psi\|^2 - 2\text{Re}(\langle \psi | \phi \rangle) + \|\phi\|^2$$

Adding:

$$\|\psi + \phi\|^2 + \|\psi - \phi\|^2 = 2\|\psi\|^2 + 2\|\phi\|^2$$

✓

1.3 The Cauchy-Schwarz Inequality

Theorem/Postulate Cauchy-Schwarz Inequality:

For all $|\psi\rangle, |\phi\rangle \in \mathcal{H}$:

$$|\langle\psi|\phi\rangle| \leq \|\psi\| \cdot \|\phi\| \quad (9)$$

Equality holds if and only if $|\psi\rangle$ and $|\phi\rangle$ are linearly dependent.

Derivation/Proof Complete Proof of Cauchy-Schwarz Inequality:

If $|\phi\rangle = 0$, both sides equal zero, and the inequality holds trivially.

Assume $|\phi\rangle \neq 0$. For any $\lambda \in \mathbb{C}$, by positive definiteness:

$$0 \leq \langle\psi - \lambda\phi|\psi - \lambda\phi\rangle$$

Expanding using sesquilinearity:

$$0 \leq \langle\psi|\psi\rangle - \lambda^* \langle\psi|\phi\rangle - \lambda \langle\phi|\psi\rangle + \lambda^* \lambda \langle\phi|\phi\rangle$$

$$0 \leq \|\psi\|^2 - \lambda^* \langle\psi|\phi\rangle - \lambda \langle\psi|\phi\rangle^* + |\lambda|^2 \|\phi\|^2$$

Choose $\lambda = \frac{\langle\psi|\phi\rangle}{\|\phi\|^2}$ (optimization parameter):

$$0 \leq \|\psi\|^2 - \frac{\langle\psi|\phi\rangle^*}{\|\phi\|^2} \langle\psi|\phi\rangle - \frac{\langle\psi|\phi\rangle}{\|\phi\|^2} \langle\psi|\phi\rangle^* + \frac{|\langle\psi|\phi\rangle|^2}{\|\phi\|^4} \|\phi\|^2$$

$$0 \leq \|\psi\|^2 - \frac{2|\langle\psi|\phi\rangle|^2}{\|\phi\|^2} + \frac{|\langle\psi|\phi\rangle|^2}{\|\phi\|^2}$$

$$0 \leq \|\psi\|^2 - \frac{|\langle\psi|\phi\rangle|^2}{\|\phi\|^2}$$

$$\frac{|\langle\psi|\phi\rangle|^2}{\|\phi\|^2} \leq \|\psi\|^2$$

$$|\langle\psi|\phi\rangle|^2 \leq \|\psi\|^2 \|\phi\|^2$$

Taking square roots:

$$|\langle\psi|\phi\rangle| \leq \|\psi\| \cdot \|\phi\|$$

Equality holds when $|\psi\rangle - \lambda|\phi\rangle = 0$, i.e., $|\psi\rangle = \lambda|\phi\rangle$. ✓

1.4 Why Complex Numbers Are Essential

Important Remark The necessity of complex numbers in quantum mechanics is not a convenience—it is a mathematical necessity rooted in the structure of interference and phase relationships.

Consider two probability amplitudes with equal magnitude but different relative phase:

$$\alpha = 1, \quad \beta = e^{i\pi} = -1, \quad \Delta\phi = \pi$$

If both real numbers were allowed:

$$|\alpha + \beta|_{\text{real}} = |1 + (-1)| = 0 \quad (\text{destructive interference})$$

This is correct! But if we restrict to magnitudes:

$$|\alpha| + |\beta| = 1 + 1 = 2 \quad (\text{wrong!})$$

The phase $e^{i\phi}$ carries information that real numbers cannot encode. More generally, any two complex numbers $\alpha, \beta \in \mathbb{C}$ can be written:

$$\alpha = |\alpha|e^{i\phi_\alpha}, \quad \beta = |\beta|e^{i\phi_\beta}$$

The relative phase $\phi_\alpha - \phi_\beta$ determines interference:

$$|\alpha + \beta|^2 = |\alpha|^2 + |\beta|^2 + 2|\alpha||\beta|\cos(\phi_\alpha - \phi_\beta)$$

This cosine term vanishes for real numbers (where $\phi = 0$ or π), making interference invisible. Complex numbers are mathematically necessary to represent quantum phenomena.

1.5 Examples of Hilbert Spaces

1.5.1 Finite-Dimensional: Spin-1/2 System

Example Space: $\mathcal{H} = \mathbb{C}^2$

Basis: The standard basis consists of:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

General State:

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

with $|c_0|^2 + |c_1|^2 = 1$ (normalization).

Inner Product:

$$\langle\psi|\phi\rangle = c_\psi^* \cdot c_\phi + d_\psi^* \cdot d_\phi$$

(if $|\phi\rangle = c_\phi |0\rangle + d_\phi |1\rangle$), or in matrix form:

$$\langle\psi|\phi\rangle = \psi^\dagger \phi = (c_\psi^*, d_\psi^*) \begin{pmatrix} c_\phi \\ d_\phi \end{pmatrix}$$

Interpretation: Spin up/down, or two-level system (qubit).

1.5.2 Infinite-Dimensional: Particle in 1D Space

Example Space: $\mathcal{H} = L^2(\mathbb{R})$, the space of square-integrable complex functions:

$$\mathcal{H} = \left\{ \psi : \mathbb{R} \rightarrow \mathbb{C} \mid \int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty \right\}$$

Inner Product:

$$\langle \psi | \phi \rangle = \int_{-\infty}^{\infty} \psi^*(x) \phi(x) dx$$

Induced Norm:

$$\|\psi\| = \sqrt{\langle \psi | \psi \rangle} = \sqrt{\int_{-\infty}^{\infty} |\psi(x)|^2 dx}$$

Normalization Condition:

$$\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

Completeness Example: Plane wave basis $\{e^{ikx}\}$ for $k \in \mathbb{R}$ in the distributional sense (requires Fourier analysis).

Physical Interpretation: Position representation of a quantum particle in one dimension. $|\psi(x)|^2 dx$ is the probability of finding the particle between x and $x + dx$.

1.5.3 Three-Dimensional Configuration Space

Example Space: $\mathcal{H} = L^2(\mathbb{R}^3)$

$$\mathcal{H} = \left\{ \psi : \mathbb{R}^3 \rightarrow \mathbb{C} \mid \int_{\mathbb{R}^3} |\psi(\mathbf{r})|^2 d^3r < \infty \right\}$$

Inner Product:

$$\langle \psi | \phi \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{r}) \phi(\mathbf{r}) d^3r$$

Norm:

$$\|\psi\| = \sqrt{\int_{\mathbb{R}^3} |\psi(\mathbf{r})|^2 d^3r}$$

Physical Interpretation: Particle in 3D space. $|\psi(\mathbf{r})|^2 d^3r$ is the probability of finding the particle in volume element d^3r centered at $\mathbf{r}$.

2 Operators and Observables: Mathematical Formalism

2.1 Linear Operators

Definition A linear operator $\hat{A}$ on Hilbert space $\mathcal{H}$ is a linear map:

$$\hat{A} : \mathcal{H} \rightarrow \mathcal{H}$$

satisfying linearity:

$$\hat{A}(\alpha |\psi\rangle + \beta |\phi\rangle) = \alpha \hat{A} |\psi\rangle + \beta \hat{A} |\phi\rangle \quad \forall \alpha, \beta \in \mathbb{C}, |\psi\rangle, |\phi\rangle \in \mathcal{H}$$

Operator Composition: For operators $\hat{A}, \hat{B}$:

$$(\hat{A}\hat{B}) |\psi\rangle = \hat{A}(\hat{B} |\psi\rangle)$$

Note: Composition is not generally commutative, i.e., $\hat{A}\hat{B} \neq \hat{B}\hat{A}$.

2.2 The Adjoint (Conjugate Transpose) Operator

Definition The adjoint (conjugate transpose) $\hat{A}^\dagger$ of operator $\hat{A}$ is defined uniquely by the relation:

$$\langle \psi | \hat{A} \phi \rangle = \langle \hat{A}^\dagger \psi | \phi \rangle \quad \forall |\psi\rangle, |\phi\rangle \in \mathcal{H}$$

In finite dimensions, $\hat{A}^\dagger = (\hat{A}^T)^* = \overline{\hat{A}^T}$ (transpose and complex conjugate).

Basic Properties:

$$(\hat{A}^\dagger)^\dagger = \hat{A} \tag{10}$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger \quad (\text{note: reversed order}) \tag{11}$$

$$(\alpha \hat{A})^\dagger = \alpha^* \hat{A}^\dagger \tag{12}$$

2.3 Hermitian Operators

Definition An operator $\hat{A}$ is **Hermitian** (or self-adjoint) if:

$$\hat{A} = \hat{A}^\dagger$$

Equivalently:

$$\langle \psi | \hat{A} \phi \rangle = \langle \hat{A} \psi | \phi \rangle \quad \forall |\psi\rangle, |\phi\rangle \in \mathcal{H}$$

Key Properties of Hermitian Operators:

1. All eigenvalues are real
2. Eigenvectors corresponding to different eigenvalues are orthogonal
3. In finite dimensions, eigenspaces form a complete orthonormal basis
4. Expectation values $\langle A \rangle = \langle \psi | \hat{A} \psi \rangle$ are always real

Derivation/Proof Proof: Hermitian operators have real eigenvalues

Let $\hat{A} |a\rangle = a |a\rangle$ be an eigenvalue equation with $\hat{A} = \hat{A}^\dagger$.

Taking the inner product with $|a\rangle$:

$$\langle a | \hat{A} a \rangle = \langle a | a \cdot a \rangle = a \langle a | a \rangle$$

Using Hermiticity:

$$\langle \hat{A} a | a \rangle = \langle a | \hat{A} a \rangle$$

But:

$$\langle \hat{A} a | a \rangle = \langle a \cdot a | a \rangle = a^* \langle a | a \rangle$$

Therefore:

$$a^* \langle a | a \rangle = a \langle a | a \rangle$$

Since $|a\rangle \neq 0$ (it's an eigenstate), $\langle a | a \rangle > 0$, so:

$$a^* = a \quad \Rightarrow \quad a \in \mathbb{R}$$

✓

Derivation/Proof Proof: Orthogonality of eigenvectors with different eigenvalues

Let $\hat{A}|a_m\rangle = a_m|a_m\rangle$ and $\hat{A}|a_n\rangle = a_n|a_n\rangle$ with $a_m \neq a_n$.

Take the inner product with $|a_m\rangle$:

$$\langle a_m | \hat{A} a_n \rangle = \langle a_m | a_n \rangle a_n = a_n \langle a_m | a_n \rangle$$

Using Hermiticity:

$$\langle \hat{A} a_m | a_n \rangle = \langle a_m | \hat{A} a_n \rangle$$

But:

$$\langle \hat{A} a_m | a_n \rangle = \langle a_m | a_m \rangle a_n = a_m^* \langle a_m | a_n \rangle = a_m \langle a_m | a_n \rangle$$

(since a_m is real for Hermitian operators)

Therefore:

$$a_m \langle a_m | a_n \rangle = a_n \langle a_m | a_n \rangle$$

$$(a_m - a_n) \langle a_m | a_n \rangle = 0$$

Since $a_m \neq a_n$:

$$\langle a_m | a_n \rangle = 0$$

The eigenvectors are orthogonal. ✓

2.4 Spectral Decomposition

Theorem/Postulate Spectral Theorem:

Any Hermitian operator $\hat{A}$ on a finite-dimensional (or suitable infinite-dimensional) Hilbert space can be written in spectral decomposition form:

$$\hat{A} = \sum_n a_n |a_n\rangle\langle a_n| = \sum_n a_n P_n$$

where:

- a_n are the distinct eigenvalues
- $|a_n\rangle$ are the corresponding orthonormal eigenstates
- $P_n = |a_n\rangle\langle a_n|$ are the projection operators onto eigenspaces
- The sum is over all eigenvalues (possibly infinite)

For a non-degenerate spectrum:

$$\hat{A} = \sum_n a_n |a_n\rangle\langle a_n|$$

For a degenerate spectrum with multiple eigenstates per eigenvalue:

$$\hat{A} = \sum_n a_n P_n \quad \text{where} \quad P_n = \sum_{i=1}^{d_n} |a_{n,i}\rangle\langle a_{n,i}|$$

(d_n is the degeneracy of eigenvalue a_n)

3 The Born Rule and Inner Products: From Geometry to Probability

3.1 The Born Rule (Fundamental Postulate)

Theorem/Postulate Postulate 1 (Born Rule):

If a quantum system is in a normalized state $|\psi\rangle$ (where $\langle\psi|\psi\rangle = 1$), and we perform a measurement of observable $\hat{A}$ with discrete spectrum, the probability of obtaining the eigenvalue a_n is:

$$P(a_n) = |\langle a_n | \psi \rangle|^2$$

where $|a_n\rangle$ is the corresponding normalized eigenstate.

For a general observable with non-degenerate spectrum:

$$P(a_n) = |\langle a_n | \psi \rangle|^2$$

For degenerate spectrum (multiple orthonormal eigenstates $\{$ where d_n is the degeneracy.

Post-measurement state: After measuring and obtaining result a_n , the state collapses (in the non-degenerate case):

$$|\psi\rangle \rightarrow |a_n\rangle$$

3.2 Geometric Interpretation

Important Remark The Born rule has a profound geometric interpretation:

$$P(a_n) = |\langle a_n | \psi \rangle|^2 = |(\text{projection of } |\psi\rangle \text{ onto } |a_n\rangle)|^2$$

By the Cauchy-Schwarz inequality, the inner product satisfies:

$$|\langle a_n | \psi \rangle| \leq \|a_n\| \cdot \|\psi\| = 1 \cdot 1 = 1$$

So the maximum probability is 1 (when $|\psi\rangle = |a_n\rangle$), and the minimum is 0 (when $|\psi\rangle$ is orthogonal to $|a_n\rangle$).

The angle θ between normalized states satisfies:

$$\cos \theta = \text{Re}(\langle a_n | \psi \rangle)$$

Thus:

$$P(a_n) = |\langle a_n | \psi \rangle|^2 = \cos^2 \theta + \sin^2 \theta \cdot (\text{phase information})$$

Quantum mechanics is literally trigonometry with consequences.

3.3 Why Magnitude Squared?

Important Remark The squaring operation in $P(a_n) = |\langle a_n | \psi \rangle|^2$ is crucial for three reasons:

(1) Non-Negativity: Probabilities must satisfy $P \geq 0$. For complex numbers $z \in \mathbb{C}$:

$$z = re^{i\theta}, \quad |z| = r, \quad |z|^2 = r^2 \geq 0$$

The quantity $|z|$ is always non-negative, and $|z|^2$ is always non-negative.

(2) Phase Independence: If the state differs by a global phase $|\psi'\rangle = e^{i\alpha} |\psi\rangle$:

$$P(a_n) = |\langle a_n | e^{i\alpha} \psi \rangle|^2 = |e^{-i\alpha}|^2 |\langle a_n | \psi \rangle|^2 = 1 \cdot |\langle a_n | \psi \rangle|^2$$

Global phase is unobservable, as it should be. This is a fundamental symmetry of quantum mechanics (U(1) gauge invariance).

(3) Distinguishing Probability from Amplitude: The magnitude squared is the operation that converts probability *amplitudes* (complex numbers) to *probabilities* (non-negative reals). This is the essence of quantum mechanics: amplitudes interfere before squaring, but probabilities do not after squaring.

Interference Formula: For two amplitudes α, β :

$$|\alpha + \beta|^2 = |\alpha|^2 + |\beta|^2 + 2\text{Re}(\alpha\beta^*)$$

The cross term $2\text{Re}(\alpha\beta^*)$ encodes interference. If we had classical probability, there would be no cross term:

$$P(\text{classical}) = P_\alpha + P_\beta$$

Quantum mechanics adds the interference term.

3.4 Proof: Total Probability Equals One

Derivation/Proof Theorem: If $|\psi\rangle$ is normalized and $\{|a_n\rangle\}$ is a complete orthonormal basis of eigenstates of $\hat{A}$, then:

$$\sum_n P(a_n) = \sum_n |\langle a_n | \psi \rangle|^2 = 1$$

Proof:

By definition of probability:

$$\sum_n P(a_n) = \sum_n |\langle a_n | \psi \rangle|^2 = \sum_n \langle \psi | a_n \rangle \langle a_n | \psi \rangle$$

Using the completeness relation (for a complete orthonormal basis):

$$\sum_n |a_n\rangle\langle a_n| = \mathbb{I}$$

We have:

$$\begin{aligned} \sum_n \langle \psi | a_n \rangle \langle a_n | \psi \rangle &= \sum_n \langle \psi | (|a_n\rangle\langle a_n|) \psi \rangle \\ &= \left\langle \psi \left| \left(\sum_n |a_n\rangle\langle a_n| \right) \psi \right\rangle = \langle \psi | \mathbb{I} \psi \rangle = \langle \psi | \psi \rangle \end{aligned}$$

By normalization, $\langle \psi | \psi \rangle = 1$, so:

$$\sum_n P(a_n) = 1$$

This confirms that probabilities sum to one, as required. ✓

3.5 Expectation Values and Uncertainties

Definition The expectation value (average measurement outcome) of observable $\hat{A}$ in state $|\psi\rangle$ is:

$$\langle A \rangle = \langle \psi | \hat{A} \psi \rangle$$

Alternatively, using the spectral decomposition:

$$\langle A \rangle = \sum_n a_n P(a_n) = \sum_n a_n |\langle a_n | \psi \rangle|^2$$

The standard deviation (uncertainty) is:

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$$

where $\langle A^2 \rangle = \langle \psi | \hat{A}^2 \psi \rangle$.

Physical Meaning: $\langle A \rangle$ is the arithmetic mean of many measurements, and ΔA is the standard deviation of the measurement outcomes.

Derivation/Proof Proof: Equivalence of expectation value formulas

Using spectral decomposition $\hat{A} = \sum_n a_n |a_n\rangle\langle a_n|$:

$$\begin{aligned} \langle A \rangle &= \langle \psi | \hat{A} \psi \rangle = \left\langle \psi \left| \sum_n a_n |a_n\rangle\langle a_n| \right| \psi \right\rangle \\ &= \sum_n a_n \langle \psi | a_n \rangle \langle a_n | \psi \rangle = \sum_n a_n |\langle a_n | \psi \rangle|^2 = \sum_n a_n P(a_n) \end{aligned}$$

This shows the two formulas are equivalent. The first is convenient for calculations, the second has intuitive meaning as a weighted average. ✓

Derivation/Proof Theorem: $\langle A \rangle$ is real for Hermitian operators

For $\hat{A} = \hat{A}^\dagger$:

$$\langle A \rangle = \langle \psi | \hat{A} \psi \rangle = \langle \hat{A} \psi | \psi \rangle = \overline{\langle \psi | \hat{A} \psi \rangle} = \overline{\langle A \rangle}$$

Therefore $\langle A \rangle = \overline{\langle A \rangle}$, which means $\langle A \rangle \in \mathbb{R}$. ✓

4 Superposition and Interference: Quantum vs Classical Probability

4.1 The Superposition Principle

Definition Postulate 2 (Linearity/Superposition):

If $|\psi_1\rangle$ and $|\psi_2\rangle$ are valid physical states, then any linear combination

$$|\psi\rangle = \alpha |\psi_1\rangle + \beta |\psi_2\rangle, \quad \alpha, \beta \in \mathbb{C}$$

is also a valid physical state (after proper normalization).

This is not an additional postulate—it follows directly from Hilbert space being a vector space.

Important Remark What Superposition Is NOT:

- Not a statement of ignorance: We don't know the state, so we're averaging
- Not rapid oscillation: The system doesn't rapidly switch between states
- Not classical probability: Classical probability of disjoint events adds linearly; quantum probabilities do not

What Superposition IS:

A superposition is a single quantum state containing multiple incompatible measurement outcomes. These outcomes are correlated through their probability amplitudes.

4.2 Probability Amplitudes vs Probabilities

Definition Amplitude: Complex number $\alpha = |\alpha|e^{i\phi}$ encoding magnitude and phase

$$\alpha \in \mathbb{C}, \quad |\alpha| \in [0, 1], \quad \phi = \arg(\alpha) \in [0, 2\pi)$$

Probability: Non-negative real number obtained by squaring the amplitude

$$P = |\alpha|^2 \in [0, 1]$$

Amplitudes undergo interference before squaring. Probabilities do not after squaring.

Crucial Difference:

Classical probability: $P(A \text{ or } B) = P(A) + P(B)$ (sum)

Quantum probability: $|\psi\rangle = \alpha|A\rangle + \beta|B\rangle \Rightarrow P(?) = |\alpha|^2 + |\beta|^2 + 2\text{Re}(\alpha\beta^*)$
(includes cross term)

4.3 Interference and the Double Slit Experiment

Example Thought Experiment: Quantum Particle Through Two Slits

A particle passes through two slits and is detected on a screen. Let $|1\rangle$ and $|2\rangle$ be the amplitudes for paths through slits 1 and 2.

Classical case (localized particle):

$$P_{\text{classical}} = P_1 + P_2$$

Leads to sum of two single-slit patterns.

Quantum case (superposition):

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle + e^{i\delta} |2\rangle)$$

where δ is the relative phase difference.

Amplitude at position on screen:

$$A_{\text{total}} = \frac{1}{\sqrt{2}}(A_1 + e^{i\delta} A_2)$$

Probability:

$$P_{\text{total}} = |A_1|^2 + |A_2|^2 + 2|A_1||A_2| \cos \delta$$

The cross term $2|A_1||A_2| \cos \delta$ produces interference fringes. The fringes vary with phase δ .

Key insight: Changing the phase δ (e.g., with a phase plate) changes the interference pattern without changing the magnitudes $|A_1|, |A_2|$. This proves phase matters.

4.4 Relative Phase and Interference

Theorem/Postulate Interference Formula:

For a superposition $|\psi\rangle = \alpha |\psi_1\rangle + \beta |\psi_2\rangle$ where $|\psi_1\rangle, |\psi_2\rangle$ are orthonormal:

$$P(\psi) = |\alpha|^2 + |\beta|^2 + 2|\alpha||\beta| \cos(\phi_\alpha - \phi_\beta)$$

where $\phi_\alpha = \arg(\alpha)$ and $\phi_\beta = \arg(\beta)$ are the phases of the amplitudes.

The interference term:

$$\Delta P = 2|\alpha||\beta| \cos(\phi_\alpha - \phi_\beta)$$

satisfies:

$$\text{Constructive: } \Delta P = 2|\alpha||\beta| \quad \text{when } \phi_\alpha - \phi_\beta = 0, 2\pi, \dots \quad (13)$$

$$\text{Destructive: } \Delta P = -2|\alpha||\beta| \quad \text{when } \phi_\alpha - \phi_\beta = \pi, 3\pi, \dots \quad (14)$$

$$\text{Partial: } \Delta P = 2|\alpha||\beta| \cos(\Delta\phi) \quad \text{otherwise} \quad (15)$$

5 Commutators and the Uncertainty Principle

5.1 Commutator Relations

Definition The commutator of operators $\hat{A}$ and $\hat{B}$ is:

$$[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A}$$

The anti-commutator is:

$$\{\hat{A}, \hat{B}\} := \hat{A}\hat{B} + \hat{B}\hat{A}$$

Basic Properties:

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}] \quad (\text{antisymmetry}) \quad (16)$$

$$[\hat{A}, \hat{A}] = 0 \quad (17)$$

$$[\hat{A}, \hat{B}] = 0 \iff \hat{A} \text{ and } \hat{B} \text{ share a complete set of eigenstates} \quad (18)$$

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}] \quad (\text{product rule}) \quad (19)$$

5.2 The Canonical Commutation Relation

Theorem/Postulate Fundamental Postulate:

For position $\hat{x}$ and momentum $\hat{p}$ operators:

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{I}$$

In higher dimensions:

$$[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij} \mathbb{I}$$

This is a fundamental postulate of quantum mechanics that cannot be derived—it defines the algebraic structure.

Important Remark The canonical commutation relation is profound:

1. **Non-zero commutator:** $[\hat{x}, \hat{p}] = i\hbar \neq 0$ means position and momentum are incompatible observables.
2. **Imaginary unit:** The factor i is essential. The commutator is not Hermitian ($[\hat{A}, \hat{B}]^\dagger = -[\hat{A}, \hat{B}]$), which is consistent with it being imaginary.
3. **Planck's constant:** $\hbar \approx 1.055 \times 10^{-34}$ J-s sets the scale. For macroscopic systems, $\hbar$ is negligibly small relative to action scales, so classicality emerges.

5.3 The General Uncertainty Principle

Theorem/Postulate Heisenberg's Uncertainty Principle (General Form):

For any two observables $\hat{A}$ and $\hat{B}$ and any state $|\psi\rangle$:

$$\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

where:

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$$

$$\Delta B = \sqrt{\langle B^2 \rangle - \langle B \rangle^2}$$

$$\langle [\hat{A}, \hat{B}] \rangle = \langle \psi | [\hat{A}, \hat{B}] | \psi \rangle$$

Derivation/Proof Complete Proof of General Uncertainty Principle:

Define centered operators:

$$\Delta A = \hat{A} - \langle A \rangle, \quad \Delta B = \hat{B} - \langle B \rangle$$

These satisfy:

$$\langle \Delta A \rangle = 0, \quad \langle (\Delta A)^2 \rangle = (\Delta A)^2$$

For any complex number $z = x + iy$:

$$|z|^2 = x^2 + y^2 > 0$$

Consider the vector state $(\Delta A + z\Delta B)|\psi\rangle$. By positive definiteness:

$$\|(\Delta A + z\Delta B)\psi\|^2 > 0$$

$$\langle \psi | (\Delta A + z \Delta B)^\dagger (\Delta A + z \Delta B) \psi \rangle \geq 0$$

Expanding using sesquilinearity:

$$\langle \psi | (\Delta A - z^* \Delta B) (\Delta A + z \Delta B) | \psi \rangle \geq 0$$

$$\langle \psi | (\Delta A)^2 \psi \rangle + z^* \langle \psi | [\Delta A, \Delta B] \psi \rangle + z \langle \psi | [\Delta B, \Delta A] \psi \rangle + |z|^2 \langle \psi | (\Delta B)^2 \psi \rangle \geq 0$$

Since $[\Delta B, \Delta A] = -[\Delta A, \Delta B]$:

$$(\Delta A)^2 + z^* \langle [\Delta A, \Delta B] | [\Delta A, \Delta B] \rangle - z^* \langle [\Delta A, \Delta B] | [\Delta A, \Delta B] \rangle^* + |z|^2 (\Delta B)^2 \geq 0$$

Note: $\langle [\Delta A, \Delta B][\Delta A, \Delta B] \rangle = \langle [\hat{A}, \hat{B}] | [\hat{A}, \hat{B}] \rangle$ since constants commute with everything.

Write $\langle [\hat{A}, \hat{B}] | [\hat{A}, \hat{B}] \rangle = c$ (a complex number). Then:

$$(\Delta A)^2 + z^*c - zc^* + |z|^2(\Delta B)^2 \geq 0$$

Choose $z = \frac{c}{(\Delta B)^2}$ to minimize over z :

$$(\Delta A)^2 + \frac{c^*c}{(\Delta B)^2} - \frac{c \cdot c^*}{(\Delta B)^2} + \frac{|c|^2}{(\Delta B)^4}(\Delta B)^2 \geq 0$$

$$(\Delta A)^2 + \frac{|c|^2}{(\Delta B)^2} - \frac{|c|^2}{(\Delta B)^2} \geq 0$$

Wait, let me redo this more carefully. Let $\langle [\hat{A}, \hat{B}] | [\hat{A}, \hat{B}] \rangle = a + ib$ where $a, b \in \mathbb{R}$.

$$(\Delta A)^2 + z^*(a + ib) - z(a - ib) + |z|^2(\Delta B)^2 \geq 0$$

Let $z = x + iy$:

5.4 Position-Momentum Uncertainty

Example Application to Position and Momentum:

From the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$:

$$\Delta x \cdot \Delta p \geq \frac{1}{2} |\langle [\hat{x}, \hat{p}] \rangle| = \frac{1}{2} |\langle i\hbar \rangle| = \frac{\hbar}{2}$$

Therefore:

$$\boxed{\Delta x \cdot \Delta p \geq \frac{\hbar}{2}}$$

This inequality is saturated by the Gaussian (minimum uncertainty) state:

$$\psi_0(x) = \left(\frac{\pi\sigma^2}{\hbar^2}\right)^{1/4} \exp\left(-\frac{x^2}{2\hbar^2/\sigma^2}\right)$$

For this state: $\Delta x = \Delta p = \sqrt{\hbar/2}$ and $\Delta x \cdot \Delta p = \hbar/2$.

Implication: To localize a particle precisely ($\Delta x \rightarrow 0$), we must have $\Delta p \rightarrow \infty$, meaning the momentum becomes completely uncertain. This is not experimental limitation—it's fundamental to reality.

6 Quantum Confinement and Energy Quantization

6.1 The Infinite Potential Well (Particle in a Box)

Definition Potential Energy Function:

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & x \leq 0 \text{ or } x \geq a \end{cases}$$

Boundary Conditions: The wavefunction must vanish at the boundaries:

$$\psi(0) = \psi(a) = 0$$

This is a hard constraint coming from the infinite potential energy: the probability amplitude cannot penetrate the barrier.

Region of Interest: We solve the Schrödinger equation only for $0 < x < a$.

6.2 Solving the Time-Independent Schrödinger Equation

Derivation/Proof Complete Solution for the Infinite Well:

Inside the well ($0 < x < a$), the time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi$$

Rearranging:

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

where $k = \frac{\sqrt{2mE}}{\hbar}$ is the wave number.

General Solution:

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

Boundary Condition at $x = 0$:

$$\psi(0) = A \sin(0) + B \cos(0) = B = 0$$

Therefore: $B = 0$ and $\psi(x) = A \sin(kx)$

Boundary Condition at $x = a$:

$$\psi(a) = A \sin(ka) = 0$$

For non-trivial solutions ($A \neq 0$):

$$\sin(ka) = 0 \implies ka = n\pi, \quad n = 1, 2, 3, \dots$$

(We exclude $n = 0$ since that gives $\psi \equiv 0$.)

Quantization Condition (from geometry):

$$k_n = \frac{n\pi}{a}$$

Energy Eigenvalues:

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2 = \boxed{\frac{n^2 \pi^2 \hbar^2}{2ma^2}}$$

Energy Eigenfunctions (unnormalized):

$$\psi_n(x) = A_n \sin\left(\frac{n\pi x}{a}\right)$$

✓

6.3 Normalization and Orthonormality

Example Normalization Procedure:

$$\int_0^a |\psi_n(x)|^2 dx = A_n^2 \int_0^a \sin^2 \left(\frac{n\pi x}{a} \right) dx = 1$$

Using the identity $\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$:

$$\begin{aligned} A_n^2 \int_0^a \frac{1 - \cos \left(\frac{2n\pi x}{a} \right)}{2} dx &= A_n^2 \left[\frac{x}{2} - \frac{a}{4n\pi} \sin \left(\frac{2n\pi x}{a} \right) \right]_0^a \\ &= A_n^2 \left[\frac{a}{2} - 0 \right] = A_n^2 \cdot \frac{a}{2} = 1 \end{aligned}$$

Therefore: $A_n = \sqrt{\frac{2}{a}}$

Normalized Eigenfunctions:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi x}{a} \right)$$

Orthonormality:

$$\langle m | n \rangle = \int_0^a \psi_m^*(x) \psi_n(x) dx = \delta_{mn}$$

where δ_{mn} is the Kronecker delta:

$$\delta_{mn} = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}$$

This orthonormality follows from the orthogonality of sine functions on the interval $[0, a]$.

6.4 Physical Interpretation: Quantization from Geometry

Important Remark The Key Insight:

Energy quantization does NOT arise from any mysterious “quantization rule.” It arises purely from: 1. Boundary conditions (hard walls) 2. Wave properties (standing waves) 3. Geometry of the box

Standing Wave Picture:

The allowed wavevectors are:

$$k_n = \frac{n\pi}{a}$$

These correspond to standing waves that fit exactly in the box:

$$a = n \cdot \frac{\lambda_n}{2} \quad \Rightarrow \quad \lambda_n = \frac{2a}{n}$$

The box must contain an integer number of half-wavelengths.

Connection to Momentum:

Using the de Broglie relation $\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p}$:

$$p_n = \frac{h}{\lambda_n} = \frac{2\pi\hbar}{\lambda_n} = \frac{2\pi\hbar}{2a/n} = \frac{n\pi\hbar}{a} = \hbar k_n$$

Energy Derivation:

$$E_n = \frac{p_n^2}{2m} = \frac{(\hbar k_n)^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{n\pi}{a} \right)^2 = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

All from geometry, no additional assumptions needed!

Ground State Energy:

The ground state has $n = 1$:

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$$

This is non-zero even though the particle is confined. This is why atoms don't collapse—quantum mechanics prevents infinite potential energy concentration.

6.5 Two-Dimensional Well

Example Separable Potential:

$$V(x, y) = \begin{cases} 0 & 0 < x < a, 0 < y < b \\ \infty & \text{elsewhere} \end{cases}$$

By separation of variables, $\psi(x, y) = X(x)Y(y)$:

$$E = E_x + E_y = \frac{n_x^2 \pi^2 \hbar^2}{2ma^2} + \frac{n_y^2 \pi^2 \hbar^2}{2mb^2}$$

$$\psi_{n_x, n_y}(x, y) = \sqrt{\frac{4}{ab}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right)$$

For a square well ($a = b = L$):

$$E_{n_x, n_y} = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2)$$

Degeneracy Example: States $(n_x, n_y) = (1, 2)$ and $(2, 1)$ have the same energy:

$$E_{1,2} = E_{2,1} = \frac{\pi^2 \hbar^2}{2mL^2} (1 + 4) = \frac{5\pi^2 \hbar^2}{2mL^2}$$

This degeneracy arises from the square symmetry. Breaking the symmetry (make $a \neq b$) lifts the degeneracy.

7 Density of States and Transport Properties

7.1 Definition of Density of States

Definition The density of states $g(E)$ is defined as:

$$g(E) = \frac{dN(E)}{dE}$$

where $N(E)$ is the cumulative number of states with energy $\leq E$.

Physically, $g(E)dE$ is the number of quantum states in the energy interval $[E, E + dE]$.

Dimensions: $[g(E)] = [\text{energy}]^{-1}$

7.2 Calculation for One-Dimensional System

Derivation/Proof Density of States for 1D Infinite Well:

Energy eigenvalues:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

Solving for n as a function of E :

$$n^2 = \frac{2mL^2 E}{\pi^2 \hbar^2} \Rightarrow n(E) = \frac{L}{\pi \hbar} \sqrt{2mE}$$

The cumulative number of states:

$$N(E) = n(E) = \frac{L}{\pi \hbar} \sqrt{2mE}$$

Density of states:

$$g(E) = \frac{dN}{dE} = \frac{L}{\pi \hbar} \frac{d}{dE} \sqrt{2mE} = \frac{L}{\pi \hbar} \sqrt{2m} \frac{1}{2\sqrt{E}}$$

$$g_1(E) = \frac{L}{2\pi \hbar} \sqrt{\frac{2m}{E}} \propto E^{-1/2}$$

Key Feature: Diverges as $E \rightarrow 0$. Very few states at low energy, many at high energy—but the density decreases with energy. ✓

7.3 Calculation for Two-Dimensional System

Derivation/Proof Density of States for 2D Square Well:

Energy eigenvalues:

$$E_{n_x, n_y} = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2)$$

In the continuum limit (large L , dense states), treat n_x, n_y as continuous variables:

$$E = \frac{\pi^2 \hbar^2}{2mL^2} n_{\perp}^2 \quad \Rightarrow \quad n_{\perp} = \frac{L}{\pi \hbar} \sqrt{2mE}$$

In 2D momentum/quantum number space, states lie in the first quadrant of a circle with radius $n_{\perp}$:

$$N(E) = \frac{1}{4} \pi n_{\perp}^2 = \frac{1}{4} \pi \left(\frac{L}{\pi \hbar} \sqrt{2mE} \right)^2 = \frac{L^2}{4\pi \hbar^2} \cdot 2mE$$

$$g_2(E) = \frac{dN}{dE} = \frac{mL^2}{2\pi \hbar^2} = \text{constant}$$

Remarkable: 2D systems have constant density of states—independent of energy!

This has profound consequences for 2D materials like graphene. ✓

7.4 Three-Dimensional System

Example Density of States for 3D Cubic Well:

For large systems, the number of states with energy $\leq E$ is the volume of an octant of a sphere in 3D momentum space:

$$N(E) = \frac{1}{8} \cdot \frac{4}{3} \pi k^3 = \frac{\pi}{6} k^3$$

where $k = \frac{\sqrt{2mE}}{\hbar}$.

$$N(E) = \frac{\pi}{6} \left(\frac{\sqrt{2mE}}{\hbar} \right)^3 = \frac{\pi}{6\hbar^3} (2m)^{3/2} E^{3/2}$$

$$g_3(E) = \frac{dN}{dE} \propto E^{1/2}$$

More states are available at higher energies.

7.5 Dimensional Dependence and Universality

Theorem/Postulate General Result for d -Dimensional Systems:

For a d -dimensional quantum system in the continuum limit:

$$g_d(E) \propto E^{d/2-1}$$

- **1D:** $g_1(E) \propto E^{-1/2}$ (diverges at $E = 0$)
- **2D:** $g_2(E) = \text{const}$ (flat)
- **3D:** $g_3(E) \propto E^{1/2}$ (increases with energy)
- **Higher:** $g_d(E) \propto E^{d/2-1}$ (increasingly steep)

This dimensional dependence has profound physical consequences:

- **1D systems:** Highly sensitive to disorder; electronic properties depend strongly on the energy window
- **2D systems:** Robust density of states; explains properties of graphene, MoS₂, and other 2D materials
- **3D systems:** Standard metals; Fermi energy well-defined

8 Tensor Products and Entanglement

8.1 Hilbert Space for Composite Systems

Definition When a quantum system consists of two (or more) subsystems, the Hilbert space is the tensor product:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$$

If $\mathcal{H}_A$ has basis $\{|a_i\rangle\}$ and $\mathcal{H}_B$ has basis $\{|b_j\rangle\}$, then $\mathcal{H}_{AB}$ has basis:

$$\{|a_i b_j\rangle\} = \{|a_i\rangle \otimes |b_j\rangle\} = \{|a_i, b_j\rangle\}$$

Tensor Product Properties:

$$\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim(\mathcal{H}_A) \times \dim(\mathcal{H}_B) \quad (20)$$

$$|\psi\rangle_A \otimes |\phi\rangle_B \neq |\psi\rangle_A + |\phi\rangle_B \quad (\text{different operations}) \quad (21)$$

$$\langle a_i b_j | a_k b_\ell \rangle = \delta_{ik} \delta_{j\ell} \quad (22)$$

8.2 Product States

Definition A state is a **product state** (or separable state) if it factorizes:

$$|\Psi\rangle_{AB} = |\psi\rangle_A \otimes |\phi\rangle_B = |\psi, \phi\rangle$$

General expansion in the basis:

$$|\psi\rangle_A = \sum_i c_i^A |a_i\rangle, \quad |\phi\rangle_B = \sum_j c_j^B |b_j\rangle$$

$$|\Psi\rangle_{AB} = \left(\sum_i c_i^A |a_i\rangle \right) \otimes \left(\sum_j c_j^B |b_j\rangle \right) = \sum_{i,j} c_i^A c_j^B |a_i b_j\rangle$$

The coefficients in the product basis form a rank-1 matrix: $C_{ij} = c_i^A c_j^B$.

8.3 Entangled States

Definition A state is **entangled** if it cannot be written as a product state. Equivalently, a state is entangled if the coefficient matrix in the product basis has rank greater than 1.

Example (Bell State):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0, 0\rangle + |1, 1\rangle)$$

In coefficient form:

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

This has rank 2, so the state is entangled.

Derivation/Proof Proof that $|\Phi^+\rangle$ is entangled:

Suppose $|\Phi^+\rangle = |\psi\rangle |\phi\rangle$ where:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\phi\rangle = \gamma |0\rangle + \delta |1\rangle$$

Then:

$$|\psi\rangle |\phi\rangle = (\alpha |0\rangle + \beta |1\rangle) \otimes (\gamma |0\rangle + \delta |1\rangle)$$

$$= \alpha\gamma |00\rangle + \alpha\delta |01\rangle + \beta\gamma |10\rangle + \beta\delta |11\rangle$$

For this to equal $|\Phi^+\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$, we need:

$$\alpha\gamma = \frac{1}{\sqrt{2}} \quad (\text{coefficient of } |00\rangle) \quad (23)$$

$$\alpha\delta = 0 \quad (\text{coefficient of } |01\rangle) \quad (24)$$

$$\beta\gamma = 0 \quad (\text{coefficient of } |10\rangle) \quad (25)$$

$$\beta\delta = \frac{1}{\sqrt{2}} \quad (\text{coefficient of } |11\rangle) \quad (26)$$

From equation (2): $\alpha = 0$ or $\delta = 0$

From equation (3): $\beta = 0$ or $\gamma = 0$

Case 1: If $\alpha = 0$, then equation (1) gives $0 = \frac{1}{\sqrt{2}}$, contradiction.

Case 2: If $\delta = 0$, then equation (4) gives $\beta \cdot 0 = \frac{1}{\sqrt{2}}$, contradiction.

Similarly for β and γ . Therefore, no factorization exists.

$|\Phi^+\rangle$ is genuinely entangled. ✓

8.4 Schmidt Decomposition

Theorem/Postulate Schmidt Decomposition Theorem:

Any pure state of a bipartite system can be written in the form:

$$|\Psi\rangle_{AB} = \sum_i \sqrt{\lambda_i} |u_i\rangle_A \otimes |v_i\rangle_B$$

where:

- λ_i are the Schmidt coefficients: $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$
- $\{|u_i\rangle\}$ form an orthonormal basis in $\mathcal{H}_A$
- $\{|v_i\rangle\}$ form an orthonormal basis in $\mathcal{H}_B$

Schmidt Rank: The number of non-zero λ_i .

Separability Criterion:

$$\text{Schmidt rank} = 1 \quad \Rightarrow \quad \text{Product state (separable)} \quad (27)$$

$$\text{Schmidt rank} > 1 \quad \Rightarrow \quad \text{Entangled state} \quad (28)$$

8.5 Entanglement Entropy

Definition The entanglement entropy of a bipartite pure state is defined as:

$$S_E = - \sum_i \lambda_i \log_2 \lambda_i$$

where the sum is over non-zero Schmidt coefficients.

Properties:

1. $S_E = 0$ for product states (separable)
2. $S_E > 0$ for entangled states
3. $S_E \leq \min(\log_2 d_A, \log_2 d_B)$ where d_A, d_B are the dimensions
4. Maximum entanglement: $S_E^{\max} = \min(\log_2 d_A, \log_2 d_B)$ (maximally entangled)

For two qubits: $S_E \leq 1$ bit, with maximum 1 bit for Bell states.

Example Entanglement Entropy Examples:**(1) Product State** $|00\rangle$: $\lambda_1 = 1$, all others = 0

$$S_E = -1 \cdot \log_2 1 - 0 - 0 - \dots = 0$$

No entanglement.

(2) Bell State $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: $\lambda_1 = \lambda_2 = \frac{1}{2}$

$$S_E = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1 \text{ bit}$$

Maximum entanglement for two qubits.

(3) Partially Entangled State $|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |11\rangle)$:

Schmidt decomposition:

 $\lambda_1 \approx 0.618$, $\lambda_2 \approx 0.382$

$$S_E = -0.618 \log_2(0.618) - 0.382 \log_2(0.382) \approx 0.95 \text{ bits}$$

Between product and maximally entangled.

9 Summary: Mathematical Framework of Quantum Mechanics

Theorem/Postulate The Five Postulates of Quantum Mechanics:

1. **State Space:** Every quantum system is described by a state vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$.
2. **Observables:** Physical observables are represented by Hermitian operators $\hat{A}$ whose eigenvalues are the possible measurement outcomes.
3. **Born Rule (Measurement):** The probability of measuring eigenvalue a_n is $P(a_n) = |\langle a_n | \psi \rangle|^2$. After measurement, the state collapses to $|a_n\rangle$.
4. **Dynamics (Schrödinger Equation):** Time evolution is governed by:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

5. **Composition:** The Hilbert space of a composite system is the tensor product of its subsystems: $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$.

All other quantum mechanics concepts follow from these five postulates and linear algebra.

This completes the rigorous mathematical foundation of quantum mechanics. All future material builds upon these concepts.

Machiraju Karthikeya